THE EFFECT OF IMAGE BLURRING ON OBJECT HALLUCINATION IN VISION-LANGUAGE MODELS

ABSTRACT

Vision-language models (VLMs) can make object-existence claims that are not supported by visual evidence. We study how Gaussian image blur affects binary object-existence predictions from Qwen2.5-VL-3B-Instruct. We evaluate the same COCO 2017 validation image-question pairs on original images and images blurred with Gaussian radius 3. In a custom COCO annotation-derived evaluation of 9,904 questions per condition, accuracy among binary answers decreases from 95.28% to 90.65%, while uncertainty increases from 1.48% to 6.14%. In the completed official POPE random split (3,000 questions per condition), accuracy decreases from 88.30% to 82.13% and recall from 77.73% to 65.40%; the false-positive rate remains 1.13%. These results suggest that blur can reduce object-existence prediction performance, particularly through additional omissions in the random split. In the completed POPE popular split, accuracy decreases from 87.47% to 80.93%, while the false-positive rate rises from 2.80% to 3.53%. The completed adversarial split shows the same recall decline and a larger false-positive-rate increase than the random and popular splits.

***Index Terms*—** Vision-language models, object hallucination, Gaussian blur, COCO, POPE, visual grounding

1. INTRODUCTION

Vision-language models (VLMs) combine visual representations with language generation to answer image questions and describe scenes. However, a model may claim that an object is present without adequate visual evidence, a failure commonly described as object hallucination. Visual degradation may make visual evidence less reliable and could change the balance between false-positive claims, missed objects, and uncertain answers.

This study examines Gaussian blur as a controlled perturbation while keeping the underlying scene and question fixed. We use Qwen2.5-VL-3B-Instruct and ask: (RQ1) How does Gaussian blurring affect object-existence prediction performance? (RQ2) Does blurring affect false-positive claims, omissions, and uncertain responses differently? We preregistered the following directional hypotheses for interpretation: H1, blur decreases object-existence accuracy and recall; H2, blur increases false-positive rate, false negatives, and/or uncertain answers; H3, the change in false-positive behavior differs by POPE negative-sampling strategy. H3 is a comparative hypothesis and is assessed descriptively here. We report a custom COCO annotation-derived evaluation and POPE random, popular, and adversarial splits; these evaluations are kept distinct. CHAIR scores are not reported because caption generation and CHAIR scoring were not performed.

2. RELATED WORK

CHAIR measures object hallucination in generated image captions by comparing mentioned objects with image annotations [1]. POPE evaluates object hallucination through binary object-existence questions and uses random, popular, and adversarial negative sampling [2]. These benchmarks offer complementary views: POPE evaluates binary object-existence responses under different negative-sampling strategies, whereas CHAIR evaluates object mentions in generated captions. Prior corruption-robustness studies often evaluate classification accuracy across clean, naturally corrupted, and adversarial inputs. Our question differs: we keep a VLM and object-existence question fixed and measure changes in yes/no grounding under a controlled visual corruption. Our custom COCO evaluation is annotation-derived and is not presented as an official implementation of either benchmark.

3. METHODOLOGY

3.1. Dataset and question construction

We use images from the COCO 2017 validation dataset and instance annotations. Each selected image contributes an annotated-present object question and an annotation-derived negative object question. This produces 9,904 image-question pairs, evaluated under both original and blurred conditions (19,808 predictions in total). The negative labels are not independently verified: an object absent from the annotations may nevertheless be visible in the image.

3.2. Visual perturbation and model

The model is Qwen2.5-VL-3B-Instruct. Each image is evaluated as originally supplied and after Gaussian blur with radius 3. The question is held constant across conditions. Model

responses are parsed as Yes, No, or Uncertain; raw answers are retained for auditing. To keep the comparison paired, the same image-question item is used in both conditions; only the image transformation changes. The evaluation pipeline records the predicted answer and applies the same label mapping and metric definitions to both conditions. This design isolates the association between the specified blur transformation and prediction changes within the selected model and benchmark setup; it does not establish causal mechanisms or generalization to other models. For POPE, each of the 3,000 questions in a split is evaluated on both the original image and its Gaussian-blurred version, yielding 6,000 predictions per split and 18,000 predictions across the three splits.

3.3. Metrics

We treat Yes as the positive class and No as the negative class. From true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), we calculate:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{3}$$

$$\text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{4}$$

$$\text{FPR} = \frac{FP}{FP + TN}. \tag{5}$$

For the custom evaluation, these binary metrics exclude Uncertain answers; uncertainty rate is U/N, where U is the number of uncertain responses and N is the total number of questions. POPE runs yielded binary answers for all questions, so all 3,000 questions per condition enter the POPE metrics. A false positive means the model answered Yes when the benchmark label was No. In the custom evaluation, a No label is derived from COCO annotations and does not guarantee that the object is visually absent; therefore, such a false positive is not necessarily a visually confirmed hallucination.

4. RESULTS: CUSTOM EVALUATION

Table 1 reports the current metrics. There are 9,904 questions in each condition. The original condition has 147 uncertain answers (1.48%), while the blurred condition has 608 (6.14%). Among binary answers, accuracy decreases by 4.62 percentage points and F1-score by 4.90 points. Recall decreases by 7.66 points, while the false-positive rate rises by 1.27 points.

The largest change among the principal classification metrics is the reduction in recall, consistent with more missed positive labels under blur. The false-positive rate also increases, and uncertain answers become more frequent. These results support the limited conclusion that blur degrades this model's object-existence performance in this setup. They do not establish that every false-positive prediction is a true hallucination because the negative labels are derived from incomplete annotations.

Table 1. Custom annotation-derived evaluation. Binary metrics exclude uncertain answers.

Metric	Original	Blurred	Change (pp)
Accuracy	95.28%	90.65%	−4.62
Precision	98.81%	97.38%	−1.44
Recall	91.70%	84.05%	−7.66
F1-score	95.13%	90.22%	−4.90
False-positive rate	1.11%	2.39%	+1.27
Uncertainty rate	1.48%	6.14%	+4.65

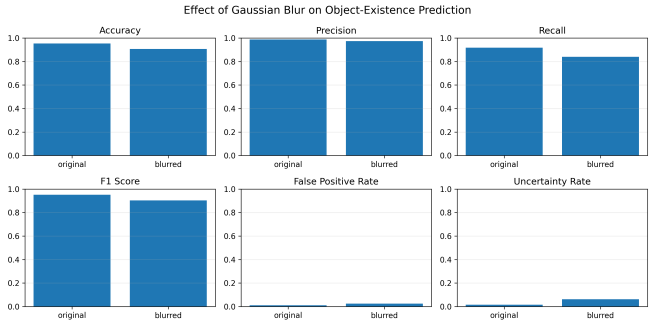


Fig. 1. Comparison of custom evaluation metrics for original and Gaussian-blurred images.

5. OFFICIAL POPE EVALUATION

5.1. Protocol

We additionally evaluated the model using the official POPE random question set (3,000 questions) and the same questions on original and Gaussian-blurred images (radius 3), yielding 6,000 predictions. The official question labels and image mapping were retained. Accuracy, precision, recall, and F1-score are computed with Yes as the positive class. False-positive rate is reported as an additional diagnostic. The random, popular, and adversarial settings are complete, with 3,000 questions per split evaluated in both conditions.

5.2. POPE random results

Table 2 summarizes the completed random split. Blur reduces accuracy by 6.17 percentage points and recall by 12.33 percentage points. The false-positive rate remains 1.13% in both conditions, while false negatives increase from 334 to 519. This pattern indicates that, in the random split, the main observed change is an increase in missed positive-labelled objects rather than an increase in false-positive predictions.

Table 2. Official POPE random split: original and Gaussian-blurred images.

Metric	Original	Blurred	Change (pp)
Accuracy	88.30%	82.13%	−6.17
Precision	98.56%	98.30%	−0.26
Recall	77.73%	65.40%	−12.33
F1-score	86.92%	78.54%	−8.38
False-positive rate	1.13%	1.13%	0.00

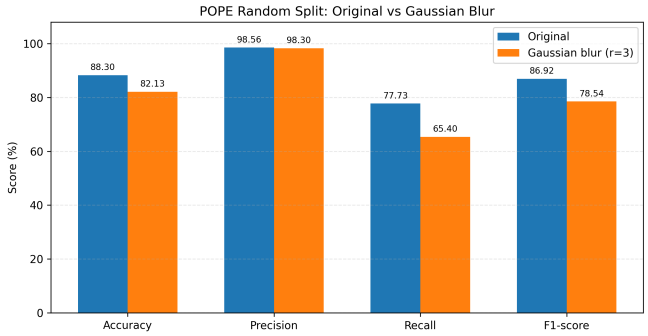


Fig. 2. POPE random split metrics for original and Gaussian-blurred images.

Table 3. Official POPE popular split: original and Gaussian-blurred images.

Metric	Original	Blurred	Change (pp)
Accuracy	87.47%	80.93%	−6.53
Precision	96.52%	94.87%	−1.65
Recall	77.73%	65.40%	−12.33
F1-score	86.12%	77.43%	−8.69
False-positive rate	2.80%	3.53%	+0.73

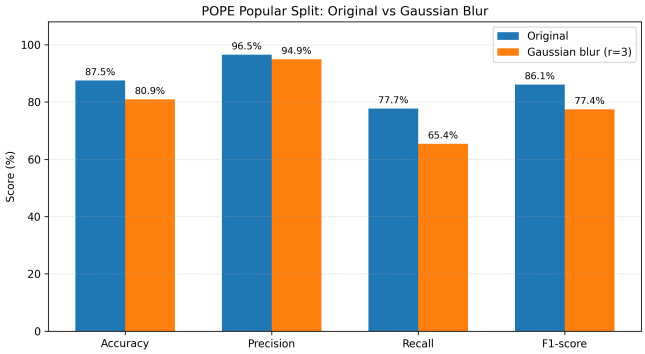


Fig. 3. POPE popular split metrics for original and Gaussian-blurred images.

The random-split result is preliminary evidence for reduced object-existence performance under blur. In that split, the false-positive rate is unchanged. The popular split shows a different pattern: accuracy decreases by 6.53 percentage points, recall decreases by 12.33 points, and the false-positive rate increases from 2.80% to 3.53%. Thus, the direction of the accuracy and recall changes is consistent across the two completed splits, while the false-positive response differs by negative-sampling strategy.

5.3. POPE popular results

Table 3 reports the completed popular split. As in the random split, blur reduces recall from 77.73% to 65.40% and increases false negatives from 334 to 519. Accuracy decreases from 87.47% to 80.93%. Unlike the random split, the false-positive count increases from 42 to 53, and the false-positive rate rises from 2.80% to 3.53%. These results indicate that the effect on false positives is not identical across POPE sampling settings.

5.4. POPE adversarial results

Table 4 reports the completed adversarial split. Accuracy decreases from 86.23% to 78.70%, recall decreases from 77.73% to 65.40%, and F1 decreases from 84.95% to 75.43%. The false-positive rate increases from 5.27% to 8.00%, with FP increasing from 79 to 120. This is consistent with H1. H2 is supported for false positives in the popular and adversarial splits, but not in the random split; the custom experiment additionally shows a higher uncertainty rate under blur. H3 receives descriptive support because the false-positive-rate change differs across splits, but we do not claim a statistically significant interaction.

5.5. Cross-split comparison

Across random, popular, and adversarial POPE splits, accuracy decreases by 6.17, 6.53, and 7.53 percentage points, respectively; recall decreases by 12.33 points in each split. False-positive-rate changes are 0.00, +0.73, and +2.73 points, respectively. This suggests that blur-related omissions are consistent across the three negative-sampling strategies, whereas false-positive changes vary by strategy. The comparison is descriptive; paired statistical tests and multiple-model replication are needed before stronger claims.

6. DISCUSSION AND LIMITATIONS

The results are consistent with the main hypothesis that Gaussian blur harms object-existence prediction. The most consistent effect in POPE is lower recall and more false negatives. The false-positive effect is conditional on the negative-sampling split: unchanged in random, but higher in popular and adversarial. The custom evaluation also shows a rise in uncertain answers from 1.48% to 6.14%.

Table 4. POPE adversarial split: original and Gaussian-blurred images.

Metric	Original	Blurred	Change (pp)
Accuracy	86.23%	78.70%	−7.53
Precision	93.65%	89.10%	−4.55
Recall	77.73%	65.40%	−12.33
F1-score	84.95%	75.43%	−9.52
False-positive rate	5.27%	8.00%	+2.73
False positives	79	120	+41
False negatives	334	519	+185

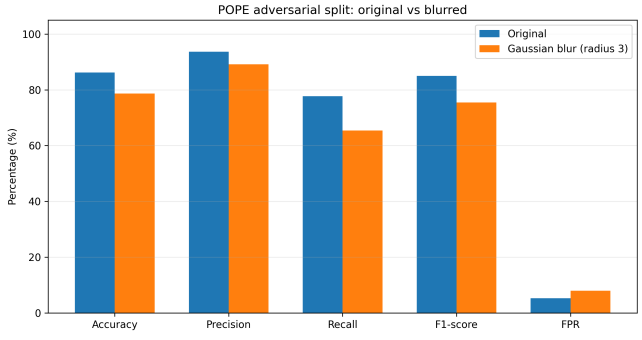


Fig. 4. POPE adversarial split metrics for original and Gaussian-blurred images.

Limitations. (1) Only one model (Qwen2.5-VL-3B-Instruct) and one blur radius (3) were evaluated. (2) No heavy-blur, occlusion, noise, or compression conditions were tested. (3) No second model or model-family replication was performed, limiting generalization. (4) The custom negative labels are derived from COCO annotations, which can omit visible objects. (5) POPE evaluates binary answers and does not directly measure free-form caption hallucination; CHAIR was not run. (6) Results are descriptive; no confidence intervals, paired significance tests, or correction for multiple comparisons are reported. (7) The exact preprocessing and inference setup should be documented and held fixed for reproducibility. (8) A model's answer change under blur may reflect visual uncertainty or altered answer bias; these experiments alone do not identify the underlying mechanism.

Follow-up hypothesis. H4: increasing blur severity causes a monotonic decline in recall, while occlusion may produce a different balance between false positives and false negatives. This is a proposed next experiment, not a tested result. A compact follow-up would use the same images/questions and model under original, mild blur, heavy blur, and one controlled occlusion condition, then report all five metrics and paired changes. A second model would improve generality but should be added only if time and compute permit. The custom experiment shows a decline in binary prediction metrics under blur and a substantial rise in uncertainty. The two completed POPE splits also show lower accuracy and recall under blur. However, the false-positive rate is unchanged in the random split and increases in the popular split, suggesting that false-positive behavior depends on the negative-sampling setting. This is a single-model experiment with one blur radius; it does not establish statistical significance, explain the causal mechanism, or demonstrate generalization to other models. The adversarial split is complete. CHAIR is not included because caption generation and annotation-based scoring were outside the completed experiments.

7. CONCLUSION

In the custom COCO annotation-derived evaluation, Gaussian blur is associated with lower binary accuracy, recall, and F1-score, alongside higher false-positive and uncertainty rates. In all three POPE splits, blur reduces accuracy and recall. The false-positive rate remains unchanged in random sampling but increases in popular and adversarial sampling. The findings support a consistent increase in omissions under this blur setting, while false-positive effects vary by sampling strategy. These findings remain limited to one model and one blur radius; further work should test multiple blur strengths, other corruption types, multiple models, paired significance tests, and caption-level CHAIR evaluation.

8. REFERENCES

[1] A. Rohrbach, L. A. Hendricks, K. Burns, T. Darrell, and K. Saenko, "Object Hallucination in Image Captioning," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018.

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